Applying Business Process Modeling Notation (BPMN) in Healthcare

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Timo Itälä M.Sc.
Aalto University

Aalto University
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Content

- In this tutorial the concepts of business processes are explained and business process modeling using the global standard notation BPMN (Business Process Modeling Notation) is introduced.
- The approach on how to identify services and design business processes based on Service Oriented Architecture (SOA) is explained.
- Concepts of data modeling related to business process and service modeling are introduced.
- A practical case study related to health care processes and integration of hospital information systems is given.

What is a Business Process?

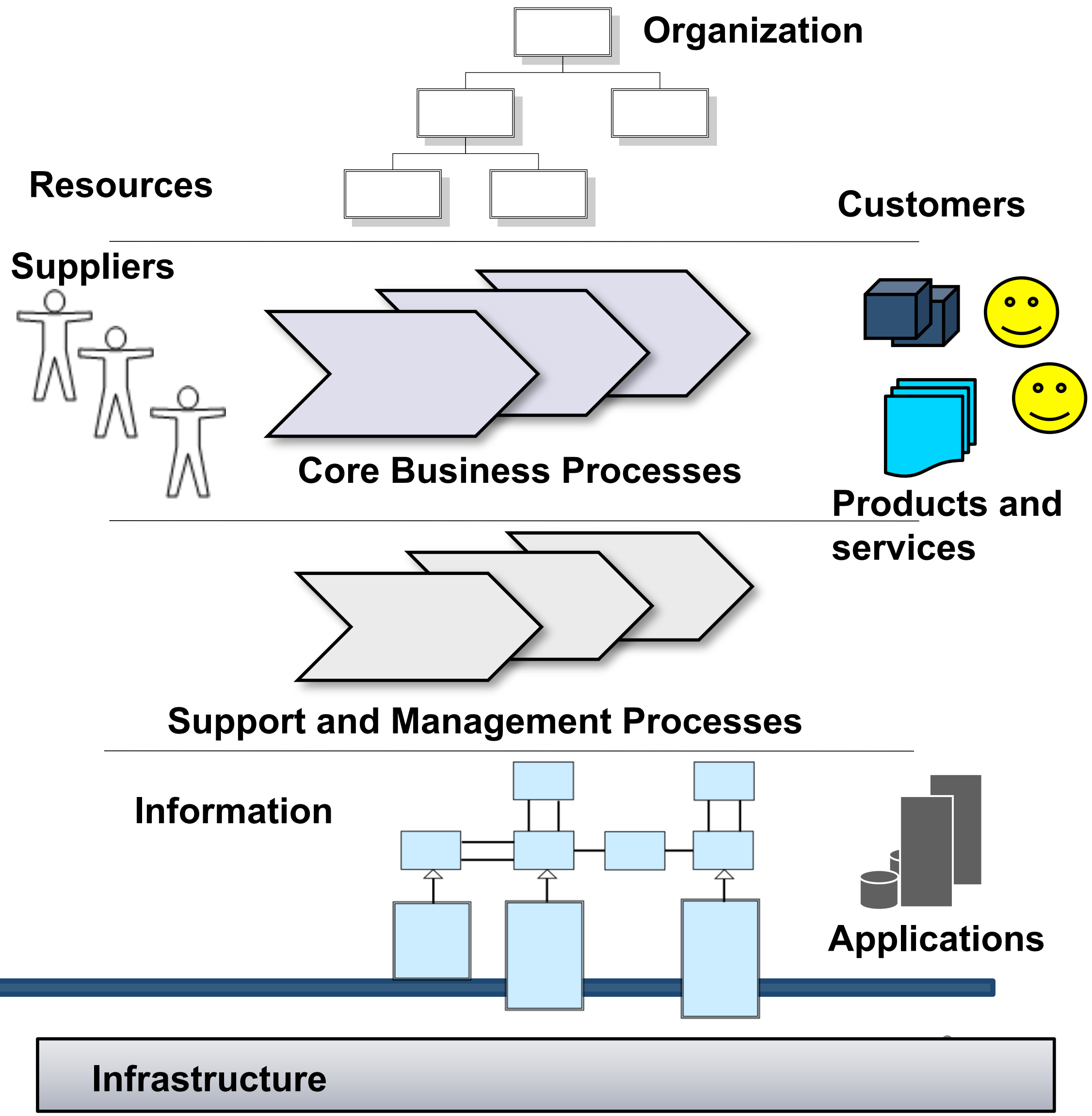
Our **organization** produces **products** and **services** for our **customers**.

The products and services are results of our **core business processes.**

Support processes give support to the core processes.

Management processes govern the operation of the system.

The processes need **information** which can be processed by our **applications** which run on our **infrastructure**.



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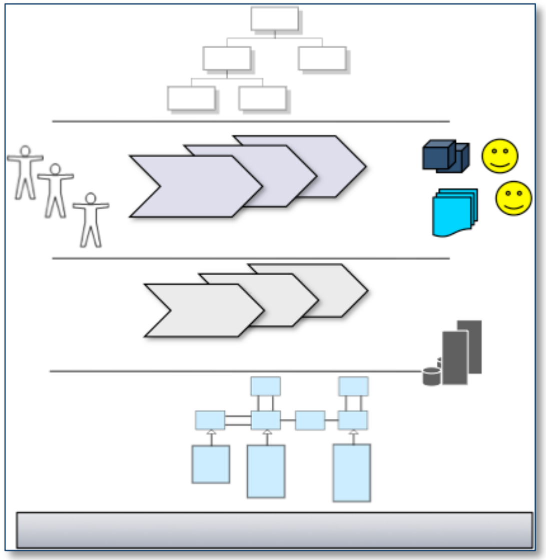
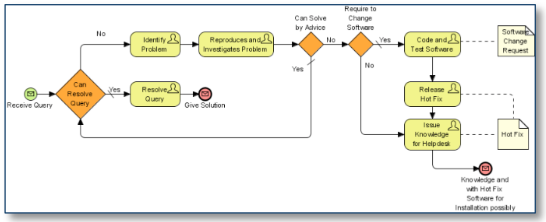
A definition of a Business Process

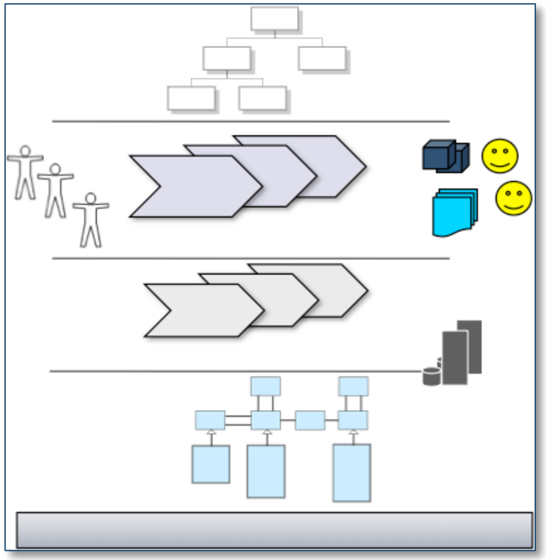
- A **business process** or **business method** is a collection of related, structured activities or **tasks** that produce a specific service or product (serve a particular goal) for a particular customer or customers. It often can be visualized with a **flowchart** as a **sequence of activities**. (Wikipedia)





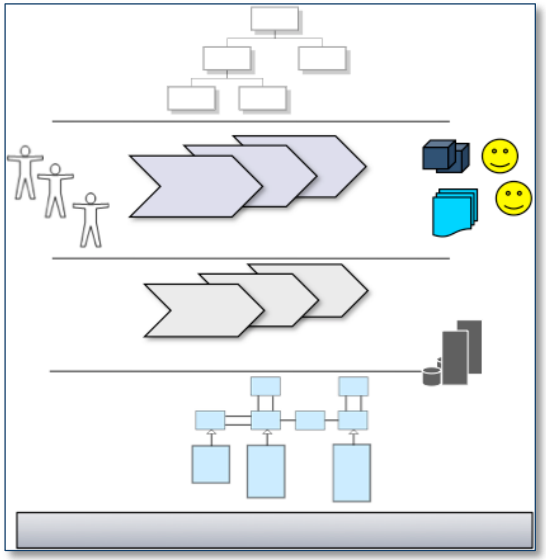
Types of processes and flows

- Real process
  - Flow of material, people, products, services etc.
- Financial process
  - Flow of money
- Information flow
  - Flow of data
- Business process
  - Flow of tasks and messages



Business processes within healthcare

- Clinical process
  - How the patient is treated
  - Diagnosis – treatment
  - Observations, tests, operations, medication
  - Executed by doctors: Give orders
- Nursing process
  - How the patient is taken care
  - Executed by nurses: Carry out orders
- Financial process
  - Collection of payments
  - The cashiers
- Scheduling process
  - Scheduling appointments
  - The assistants
- Patient record process
  - Doctors, Patients, other staff

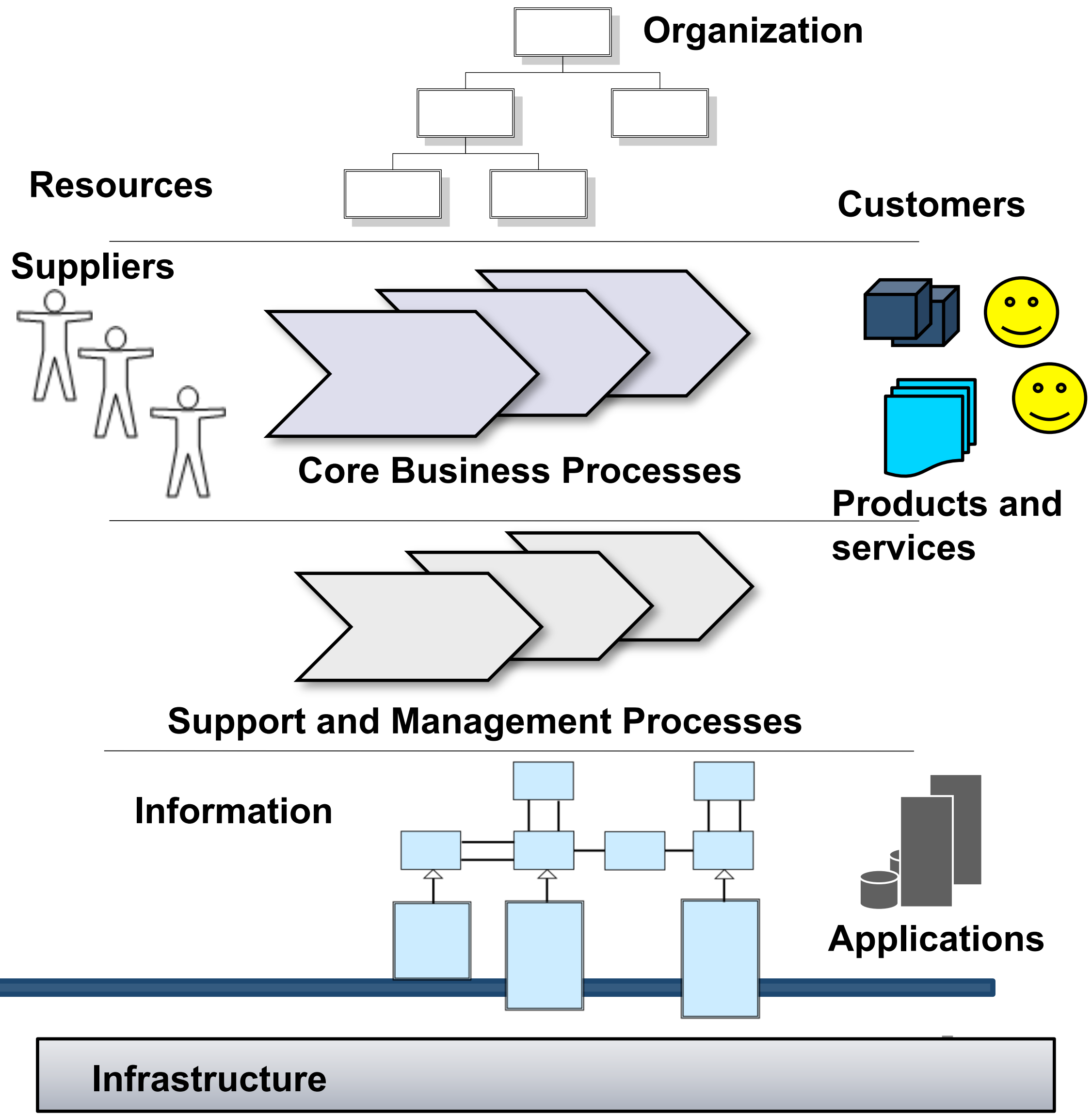


A Model and an Instance of a Business Process

A Model of a business process describes how product and services are produced

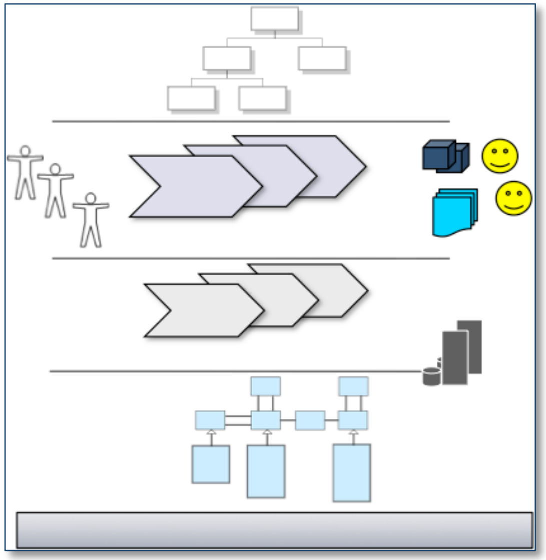
An Instance of a business process produces an uniquely identifiable product or service to an uniquely identifiable customer

Business process modeling and information modeling go hand-in-hand



Why model business processes?

If we want to improve how we do things we must first understand how we do things today



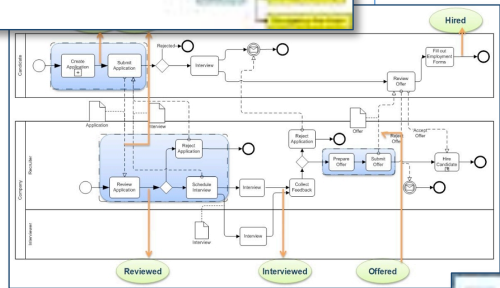
- First we model as-is
- Then we look at the problems or opportunities for improvement
- Then we choose the most important improvements
- Then we model the to-be

Modeling business processes

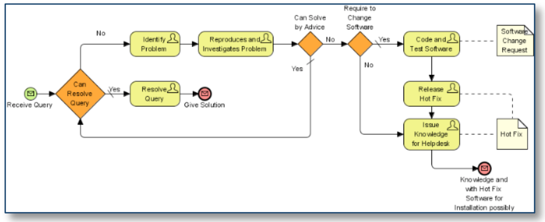
How to model and represent business processes?

- Verbal, textual descriptions
- Visual diagrams
- Execution instructions

1. Starting with a customer placing an order **(the customer need)**
2. send IT-based information to the warehouse
3. stock picking
4. packing and recording
5. sending the appropriate IT-based information to the distribution hub
6. sending IT-based information to the accounts department
7. generation of an invoice
8. allocation and organisation of shipment for the vehicle drivers
9. delivery of the item and invoicing **(the customer need fulfilled).**

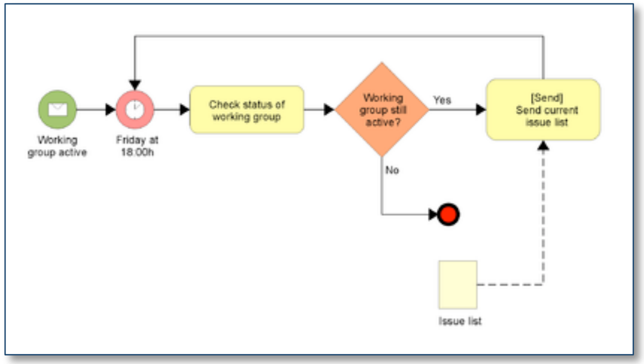



```
<process name="EMailVotingProcess">
  <!-- The Process data is defined first-->
  <sequence>
    <receive partnerLink="Internal" portType="tns:processPort"
             operation="receiveIssueList" variable="processData"
             createInstance="Yes"/>
    <invoke name="ReviewIssueList" partnerLink="Internal"
            portType="tns:internalPort" operation="sendIssueList"
            inputVariable="processData" outputVariable="processData"/>
    <switch name="Anyissuesready">
      <!-- name="Yes" -->
      <case condition="bpws:getVariableProperty(ProcessData,NumIssues)>0">
         <invoke name="DiscussionCycle" partnerLink="Internal"
                 portType="tns:processPort" operation="callDiscussionCycle"
                 inputVariable="processData"/>
         <!- Other Activities not shown -->
      <!--name="No" -->
      </case>
      <otherwise>
        <empty/>
      </otherwise>
    </switch>
  </sequence>
</process>
```



Business Process Modeling Notation (BPMN)

- BPMN is a grphical representation for specifying business processes in a workflow
- BPMN was developed by Business Process Management Initiative (BPMI)
- BPMN is currently maintained by the Object Management Group (OMG) since 2005
- BPMN 2.0 published recently
- Tool support: (62 listed)
  - Drawing tools
  - Repository based modeling tools



Level of Detail?

Data, Messages

Processes, Services, Operations

Choreography

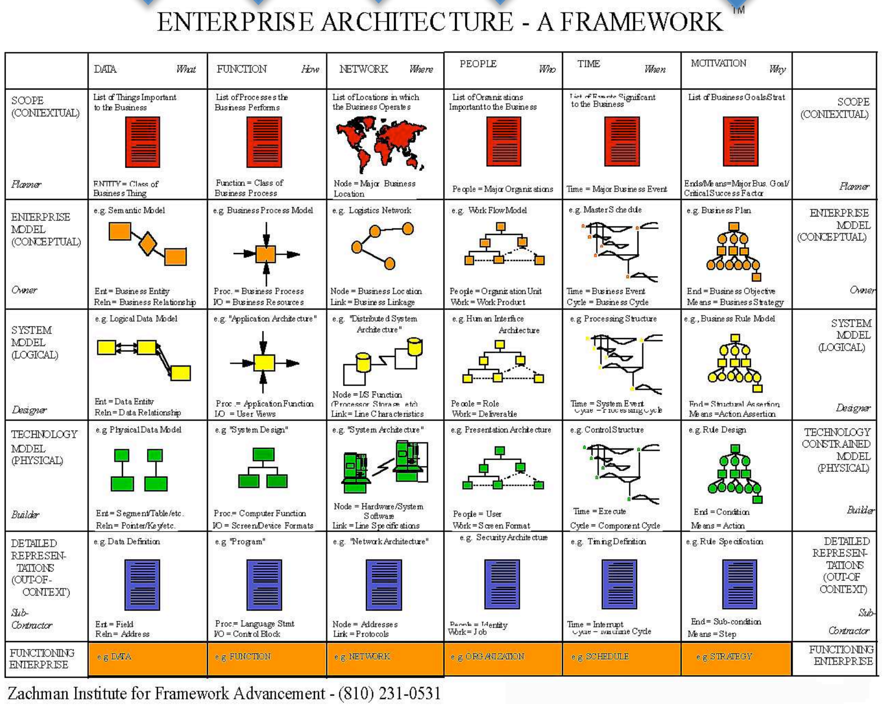
Pools, Lanes, Partners

Problems, Solutions

Level 1: Conceptual, Descriptive

Level 2: Logical, Analytical

Level 3: Physical, Executable

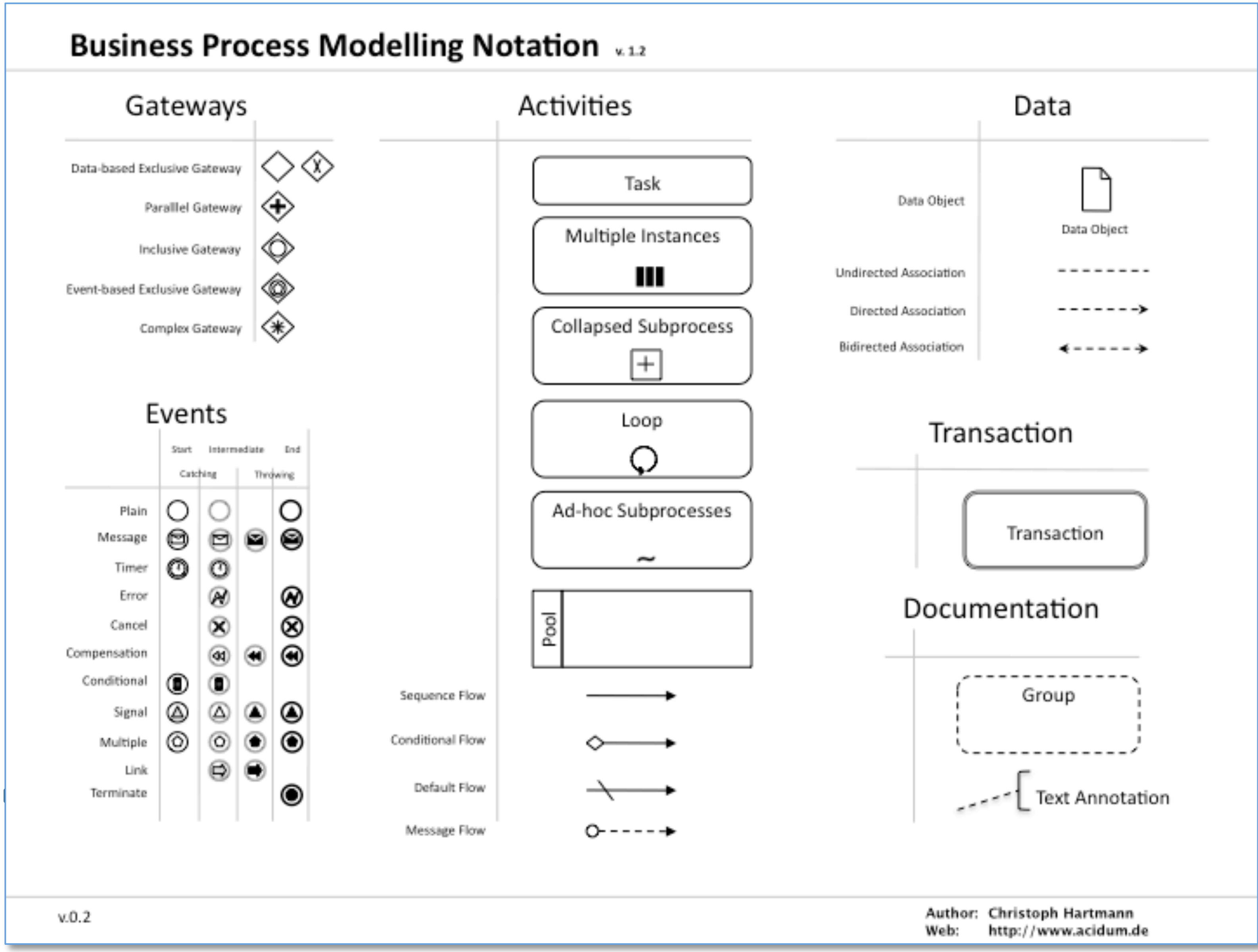


Business Process Redesign

- Effectiveness
  - To do the right things
- Efficiency
  - To do things right

http://apps.adcom.uci.edu/EnterpriseArch/Zachman/zachman.jpg

BPMN Elements

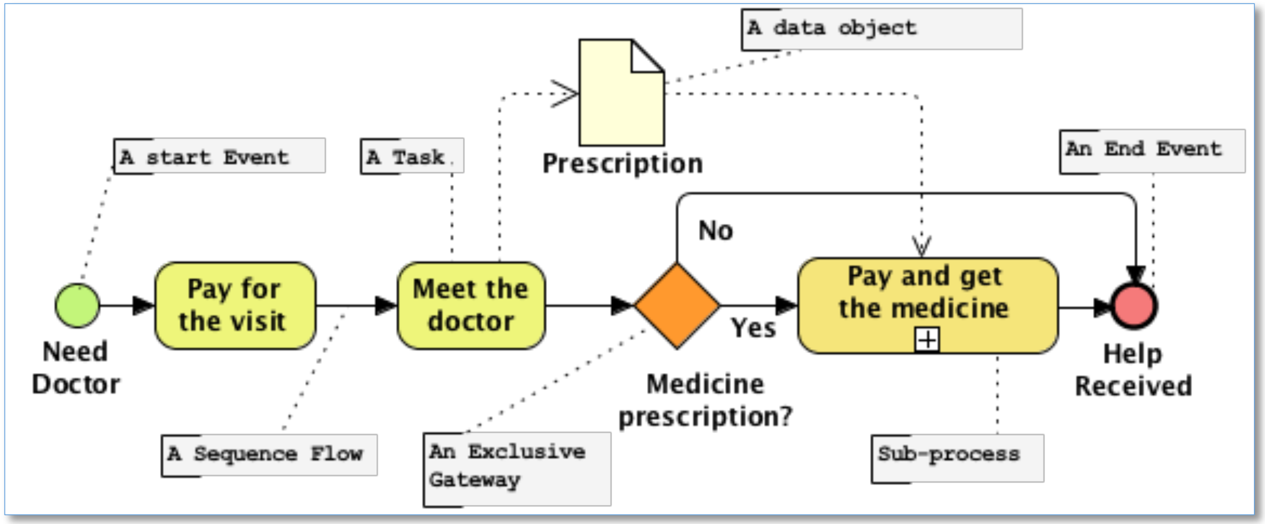


Example: The Patient in the "Happy Hospital"

Business Process Steps

- Pay for the book, if you don´t have one
- Pay for the visit
- Choose the department
- Wait in the line
- Meet the doctor
- Go to the lab
- Pay for the test
- Give the sample
- Get the results form
- Wait in the line
- Meet the doctor again
- The doctor writes the observations and the medication in the book
- Pay for the medicine
- Get the medicine

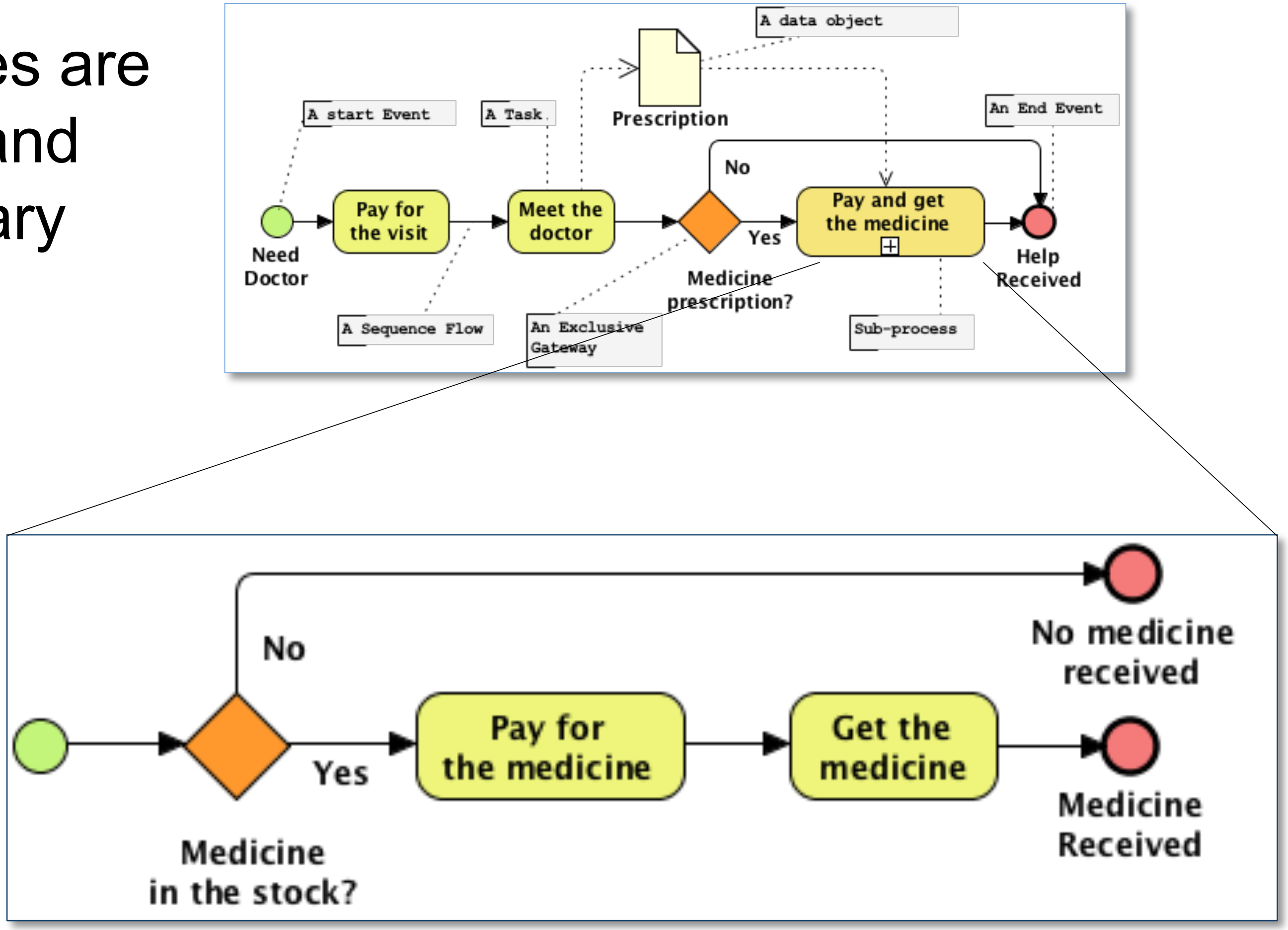
Example: Hospital visit as a BPMN diagram



- Flow objects: Events, Tasks, Gateways
- Connecting Objects: Sequence Flow, Annotation
- Data Objects: Data and Documents in the Process
- Hiding details: Sub-Process

Hiding details: Sub-Process

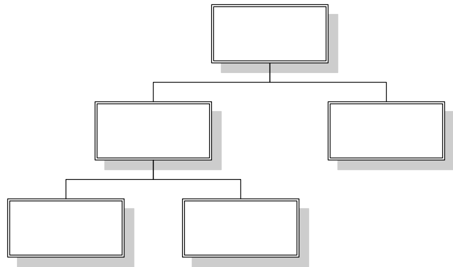
Sub-Processes are used to hide and show necessary level of detail



Participants in the processes

Participants

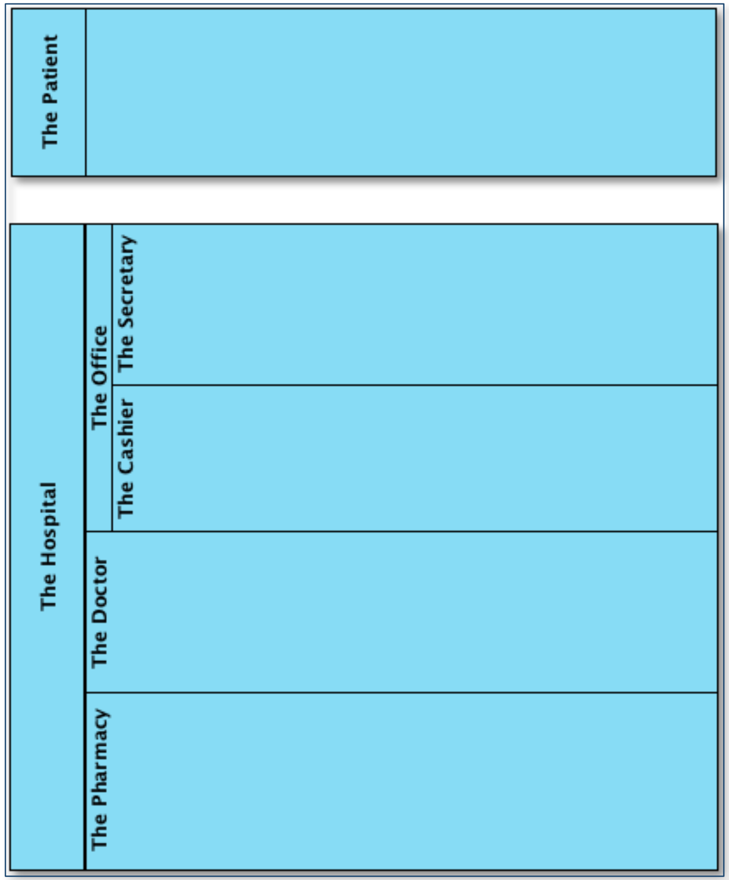
- Organizations or departments
  - The Hospital
  - The Pharmacy
- Roles of persons
  - The Patient
  - The Doctor
  - The Nurse
  - The Receptionist
- How do we represent participants in the processes?



Participants: Swimlanes, Pools and Lanes

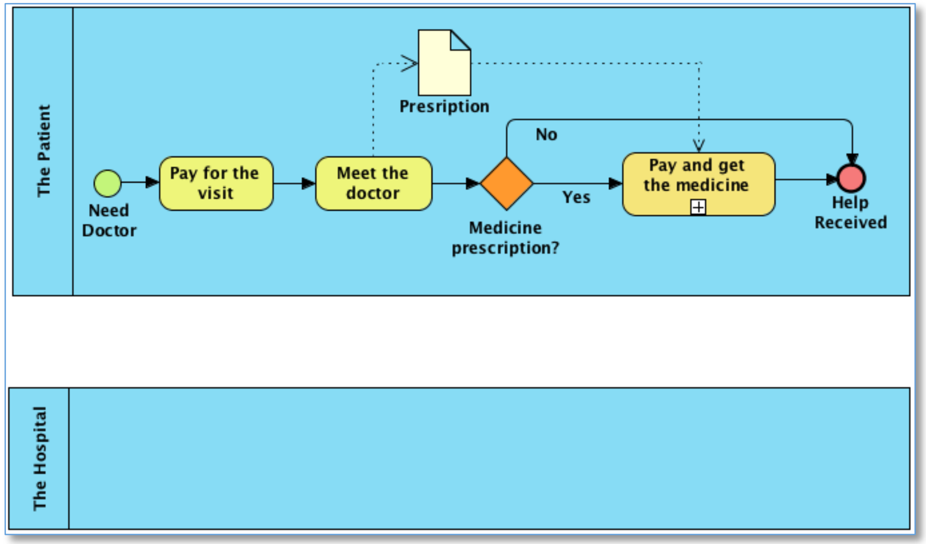
- A Pool represents a process of one participant
- A Pool can contain sub-partitions to show different roles within a participant

- Correction: Assistant !



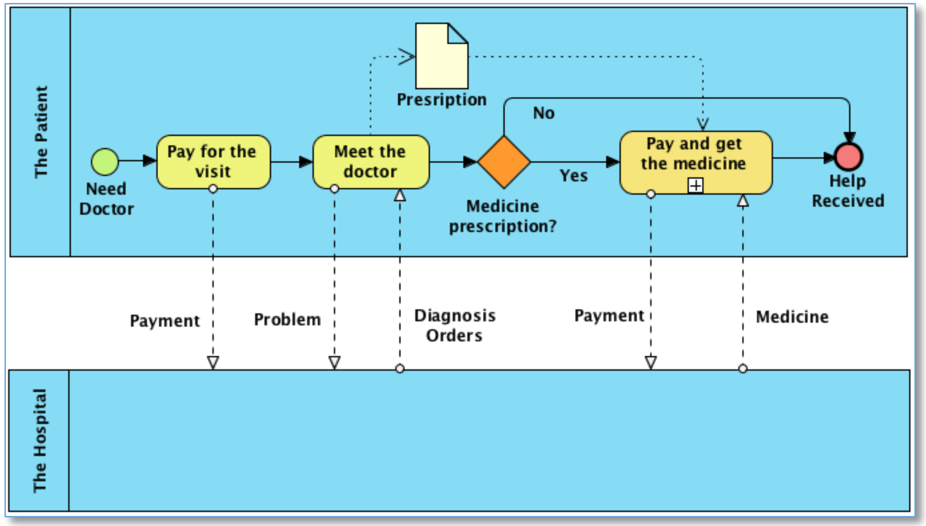
Processes within Pools

- A Business Process is always within one Pool
- The Patient Pool: White-Box pool
- The Hospital Pool: Black-Box Pool
- Participants and their processes can collaborate with each other. How?



Collaboration with messages between processes

- The participant´s processes can collaborate using messages
- Message flow always between pools
- Sequence flow always within a pool



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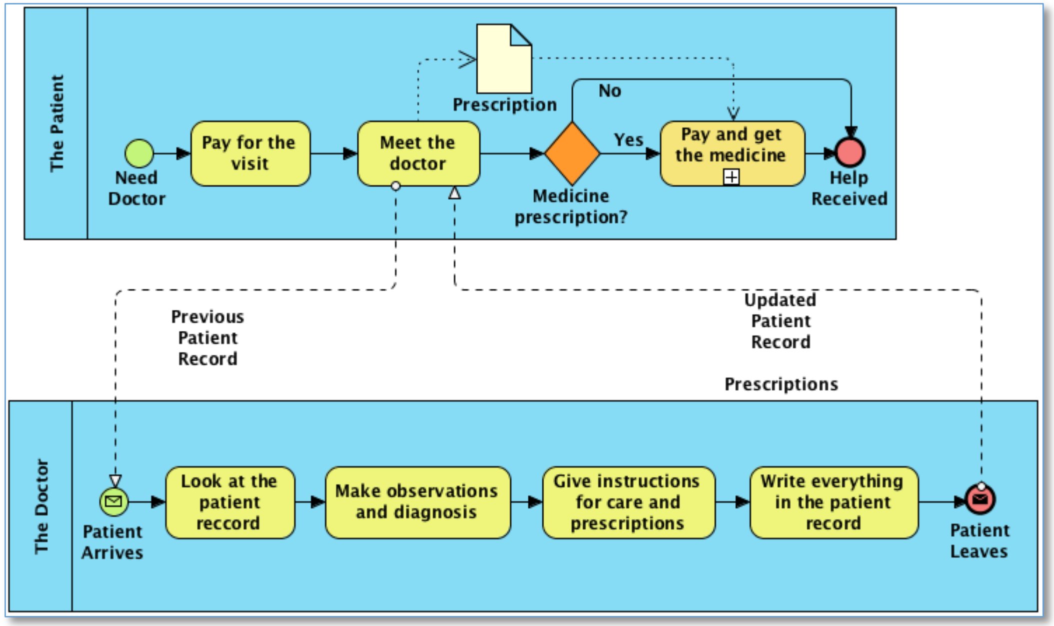
The Core Process of an organization

Core Process characteristics

- The doctor´s process gives a service to the patient
- Volumes in Happy Hospital
  - 1000 visits/day
  - 600 beds
  - 10 000 employees



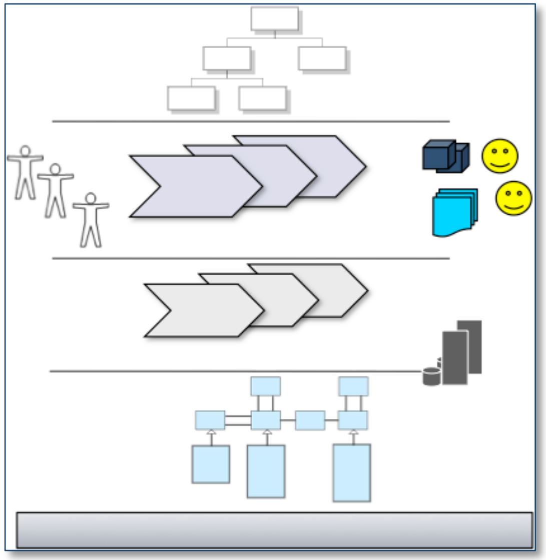
Business Process Modeling Method steps

1. Define Process Scope
2. Create the Top Level diagram for the Happy Path
3. Add top-level exception paths
4. Expand sub-processes to show detail at child level
5. Add intermediate message flows to external pools

Discovering the processes

- What are the core processes?
- What are the support processes?

- Service orientation
  - Support processes give services to the core processes



What to improve? Process Orientation

- Clinical Process?
- Scheduling Process?
- Financial Process?
- Pharmacy Process?
- Other Processes?
- Health Record Management Process?
  - As-Is: Health Records are written in the book owned by the patient
  - Problems: A book can be in one place only. Other problems...

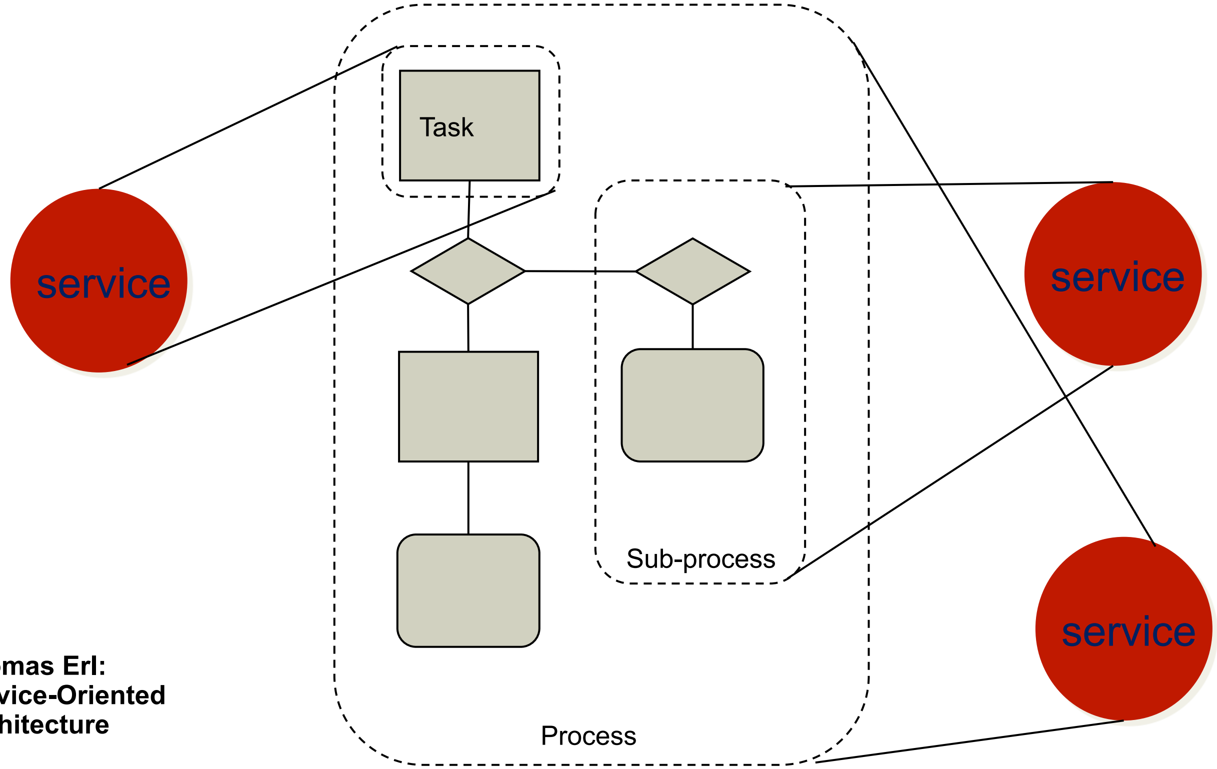
How to improve? Service Orientation

- Business Processes are composed of business services
  - Examples: Payment service, Patient record service
- Business Services are reusable components which can be used in many business processess
- Business Services can be implemented using software components, often web services
- The Benefits of using reusable components
  - Cost savings: Build once, use many times
  - Time savings: Use ready components instead of building from scratch
  - Risk management: Using ready components helps to control the risks

Discovering services

SOA

Retrieve
Patient
Data



Pay and get
the medicine

Make the
Laboratory
tests

Thomas Erl:
Service-Oriented
Architecture

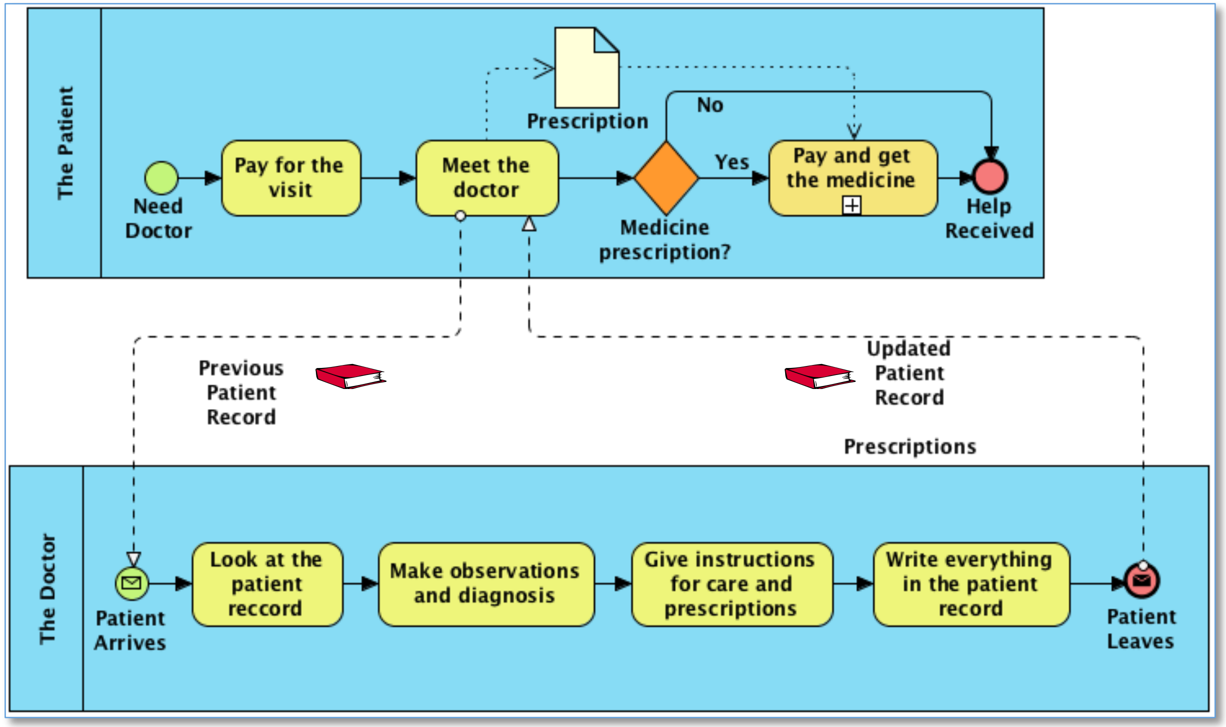
A service can be a task, a sub-process or a process

From As-Is to To-Be process

- Discussion about services
  - How to identify services
  - SOA Principles
- How the processes and services could be identified
  - Data oriented services (patient record)
  - Function oriented services (laboratory)
  - Process oriented services (the doctors workstation)
  - Notification oriented services (do we have any?..)

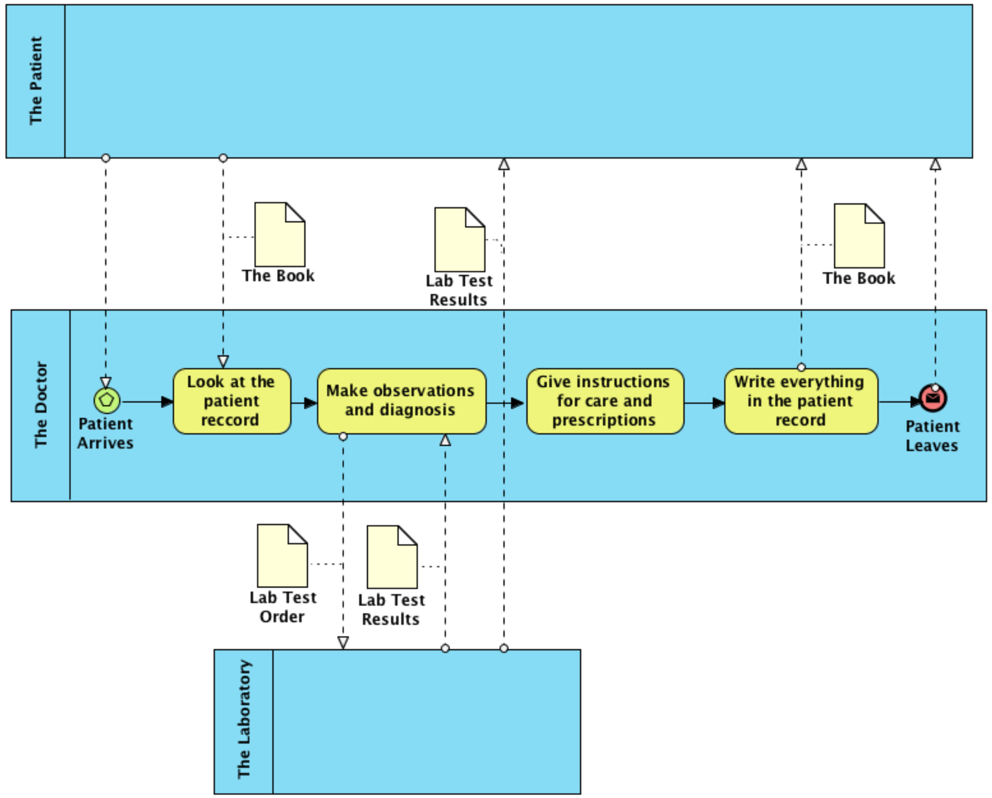
The Patient Record and the Doctor

- The patient owns the book



Laboratory included as a business service

- The Laboratory orders and results are on a separate paper form
- The doctor receives the results
- The patient pays for the laboratory and receives the results

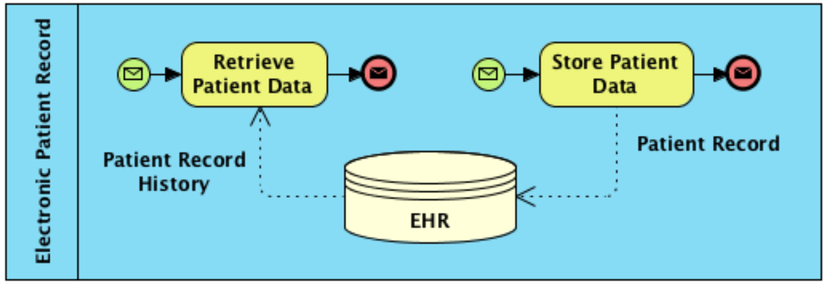


Ideas for improvement regarding Patient Health Record

- Electronic Health Record?
- Discussion and collection of improvement ideas
  - What problems it could solve?
    - The patient could forget the paper book at home or lose it
    - The doctor has a bad handwriting
    - The laboratory results and other documents are on separate papers
  - What other opportunities EHR would give
    - The hospitals could share the patient records
- Other requirements
  - Privacy, Confidentiality, Authenticity and other security aspects
  - Support to other processes: Scheduling, Financial, etc...
  - Availability, Usability, Performance

Electronic Health Record as a service

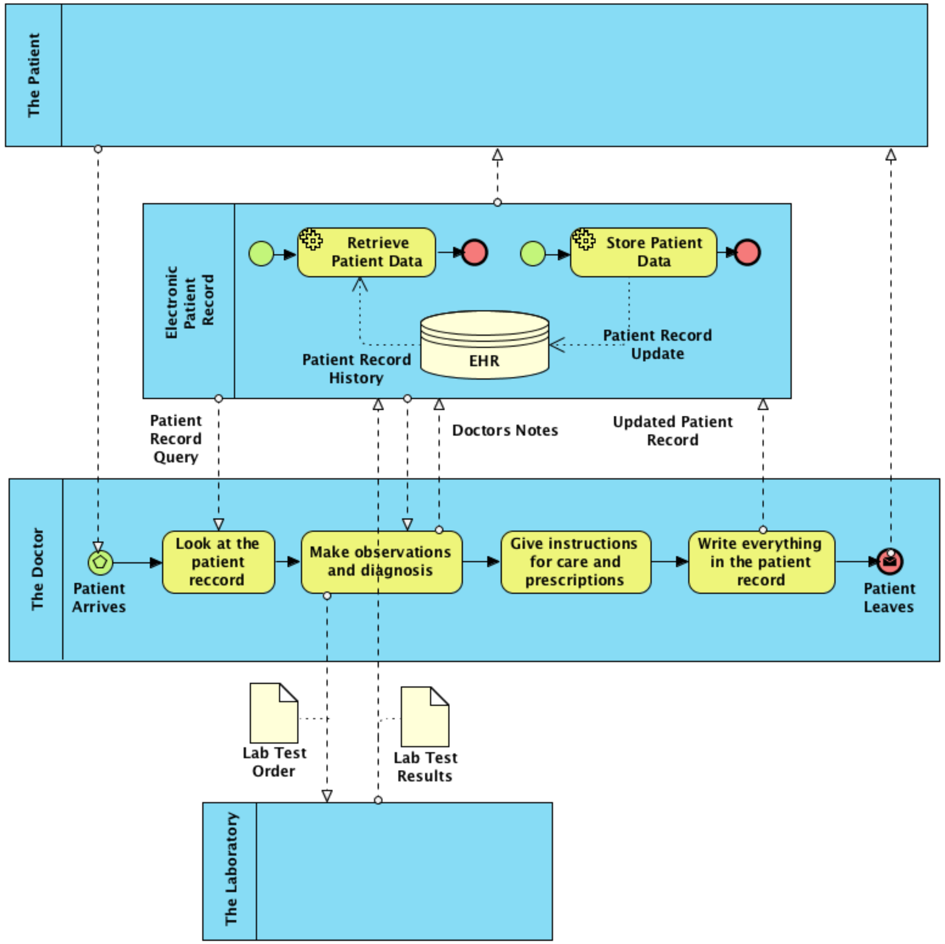
- EHR could store all the doctors notes in a similar way as the patient record book
- It could store also the lab results and other information
- It could be accessible for the doctor and other professionals when needed
- It could also be accessible within the hospital and also outside the hospital like regional level



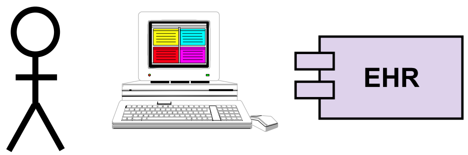
- It must guarantee the privacy, confidentiality and authenticity of the notes

How EHR service could be used?

- The doctor would start looking at the patient´s EHR
- The lab results would be collected into the EHR
- The doctor would write all notes into the EHR
- The patient would get a paper copy or could also look at the EHR

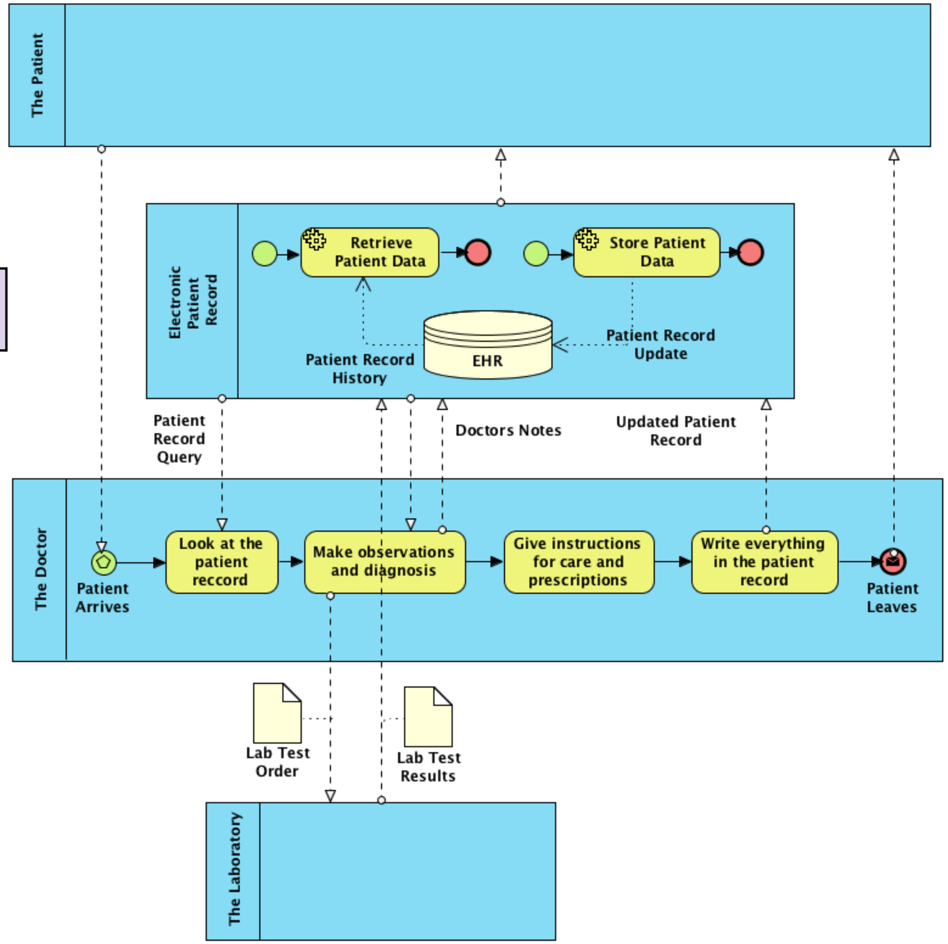


How EHR could be implemented?



Option 1

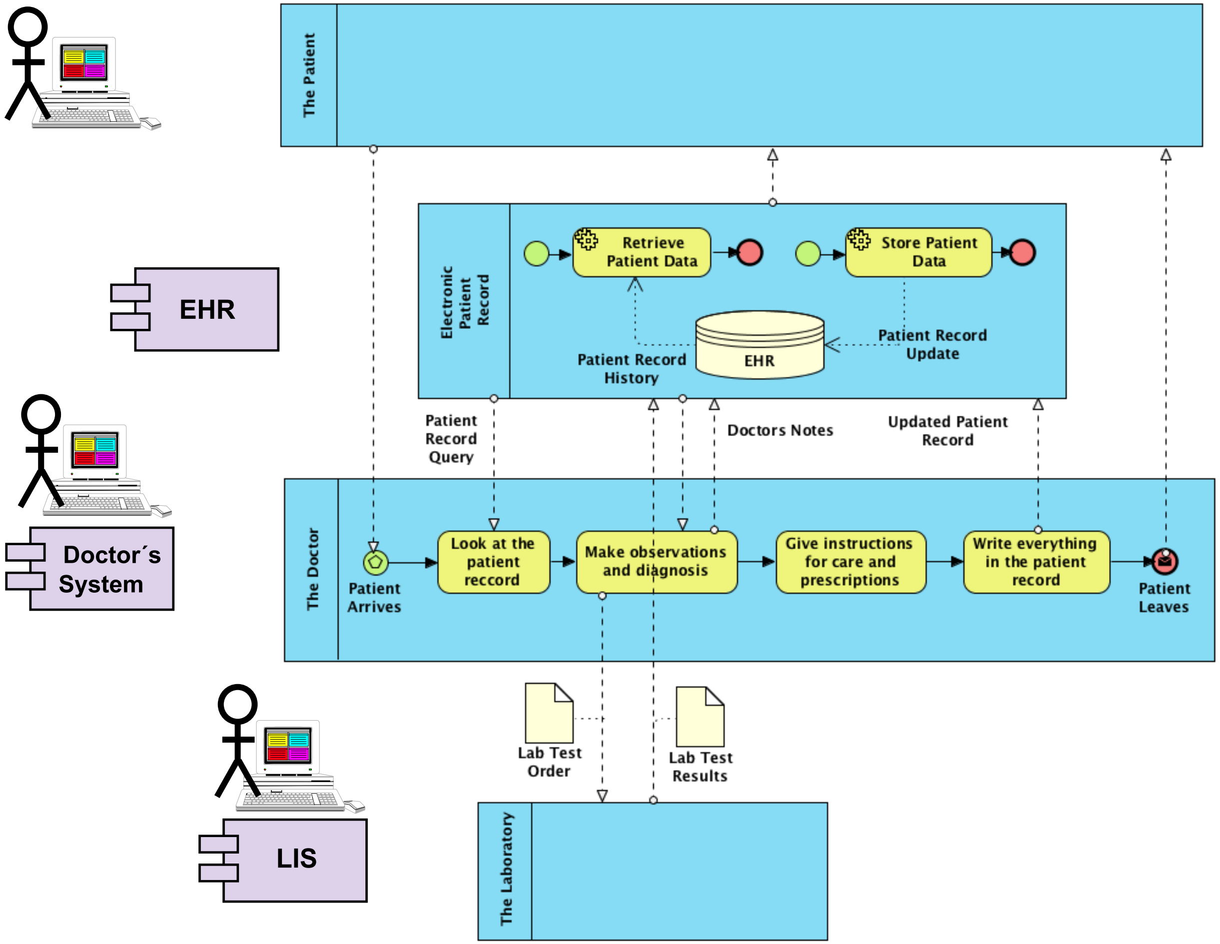
Stand alone system



How EHR could be implemented?

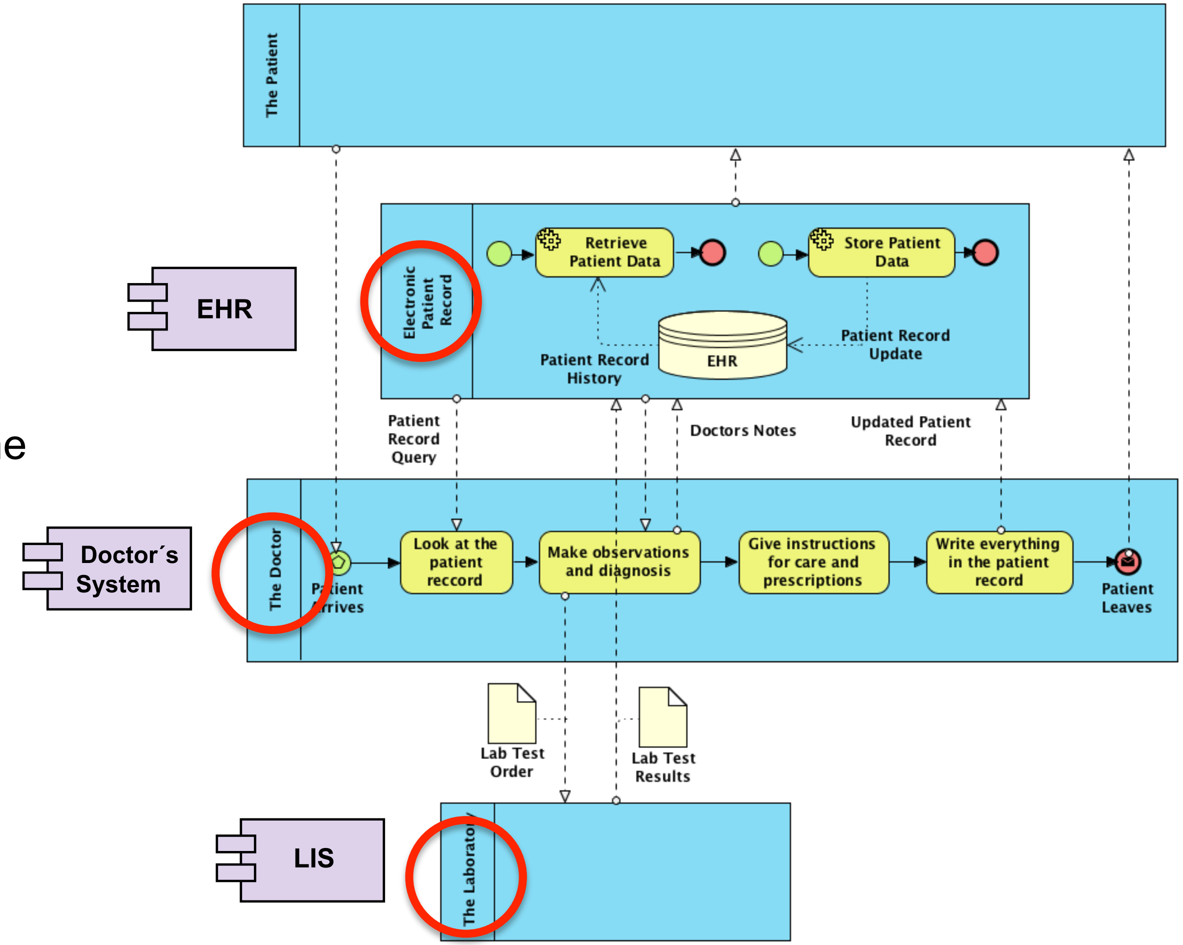
Option 2

An application service which would offer services to other applications



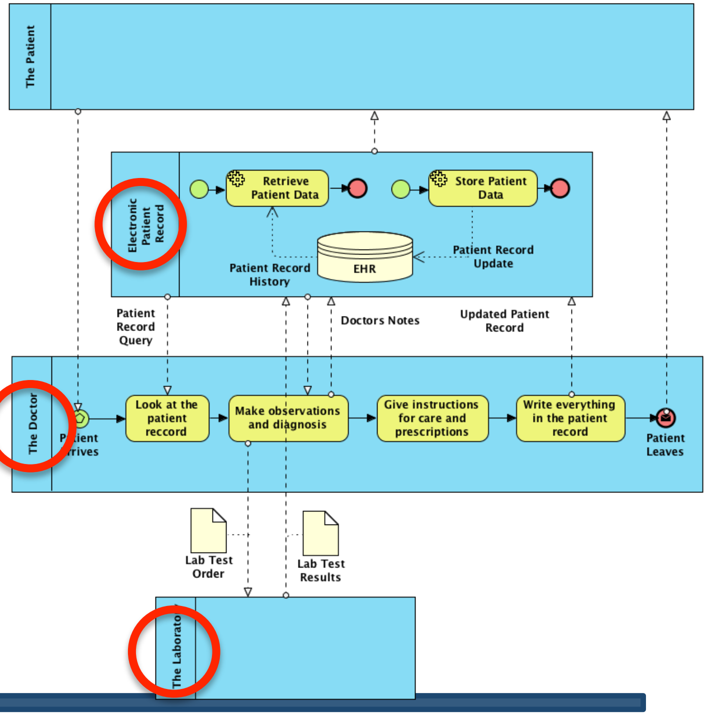
Identifying Application Services

- EHR
  - Patient record management
- LIS
  - Laboratory order entry and results rdelivery
- Pharmacy system
  - Delivery of the medicine
- Doctor´s systems
  - Coordination of the collaboration
- The Patient
  - Collaboration with the professionals



Implementing services as web services

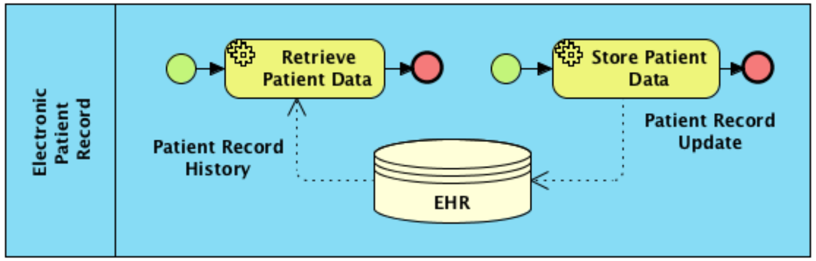
- A **service** consists of one or multiple **operations**
  - A Message Exchange Pattern (MEP) is related to an operation
  - **Request-Response** operation
    - A service receives a request message and sends a reply message
  - **Solicit-Response** operation
    - A service sends a request message and waits for a reply message
  - **One-way** operation
    - A service receives a message
  - **Notification** operation
    - A service sends a message
  - A fault message can be replied (Fault)
- Applies to any programming language
- Services are **synchronous** or **asynchronous**



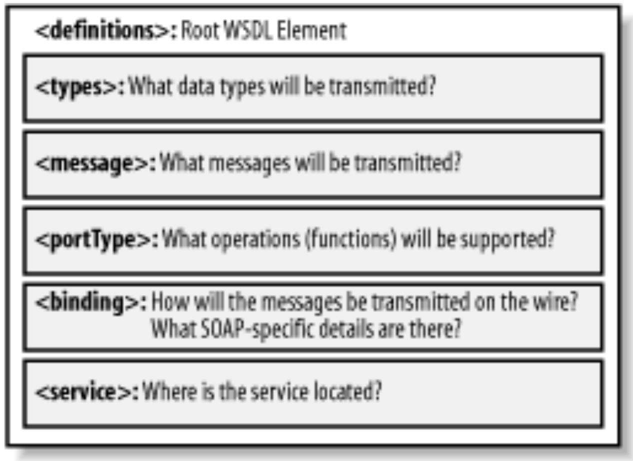
Implementing services as web services

Service: Electronic Patient Record

- Operations and messages
  - EPRQuery
    - In: EPR-QueryMessage
    - Out: EPR-ReplyMessage
  - EPRStore
    - In: EPR-StoreMessage

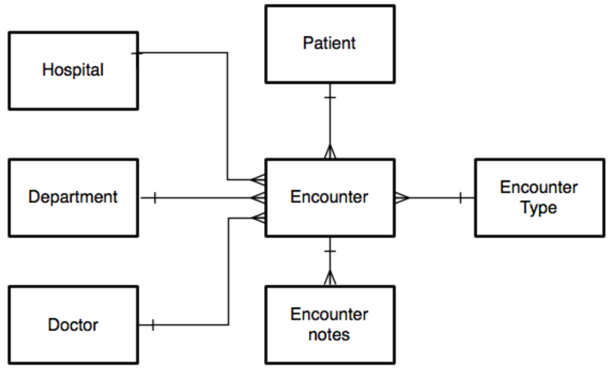


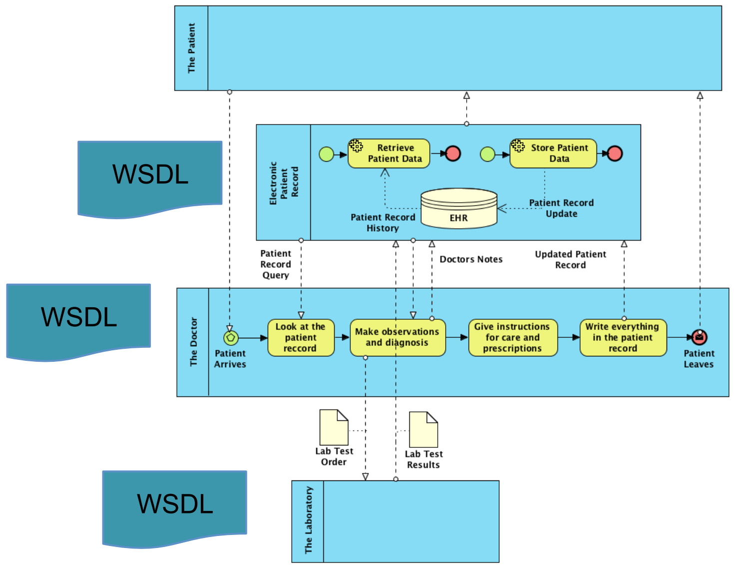
- Web services are defined using web services definition language (WSDL)



The data model for the messages

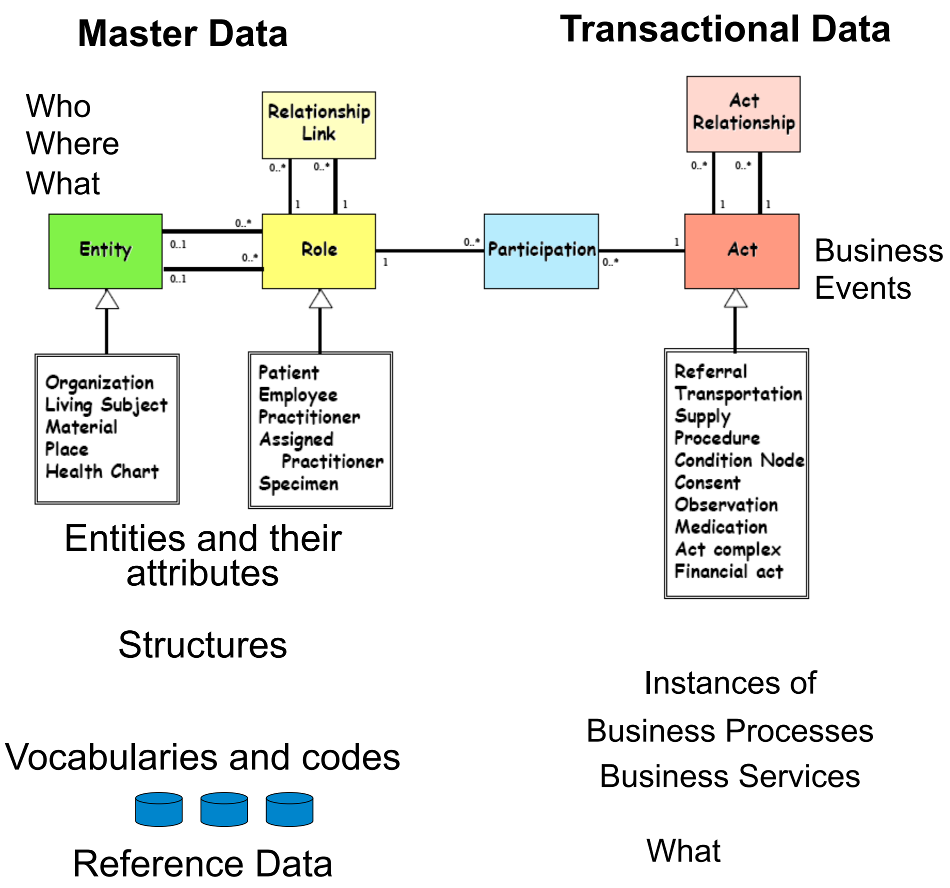
- How do we define the messages in WSDL documents?
- We need a data model
- Transactional data
  - What happened
  - Encounter and Notes...
- Master Data
  - The "static data" that is referenced from the transactional data that describes business events
  - Hospital, Department, Doctor, Patient, Encounter Type...





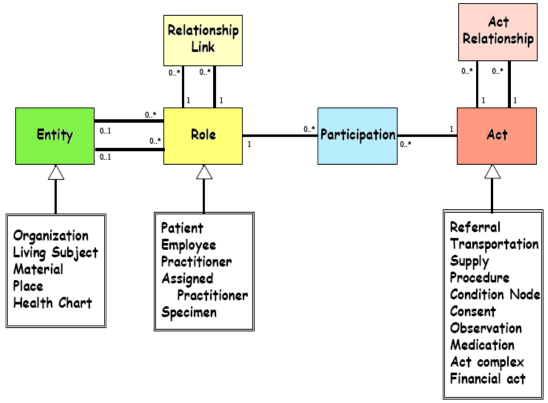
Healthcare Information Model HL7 RIM

- RIM (Reference Information Model) is a generic health care data model
- HL7 CDA (Clinical Document Architecture) is a RIM based standard for exchange of clinical records



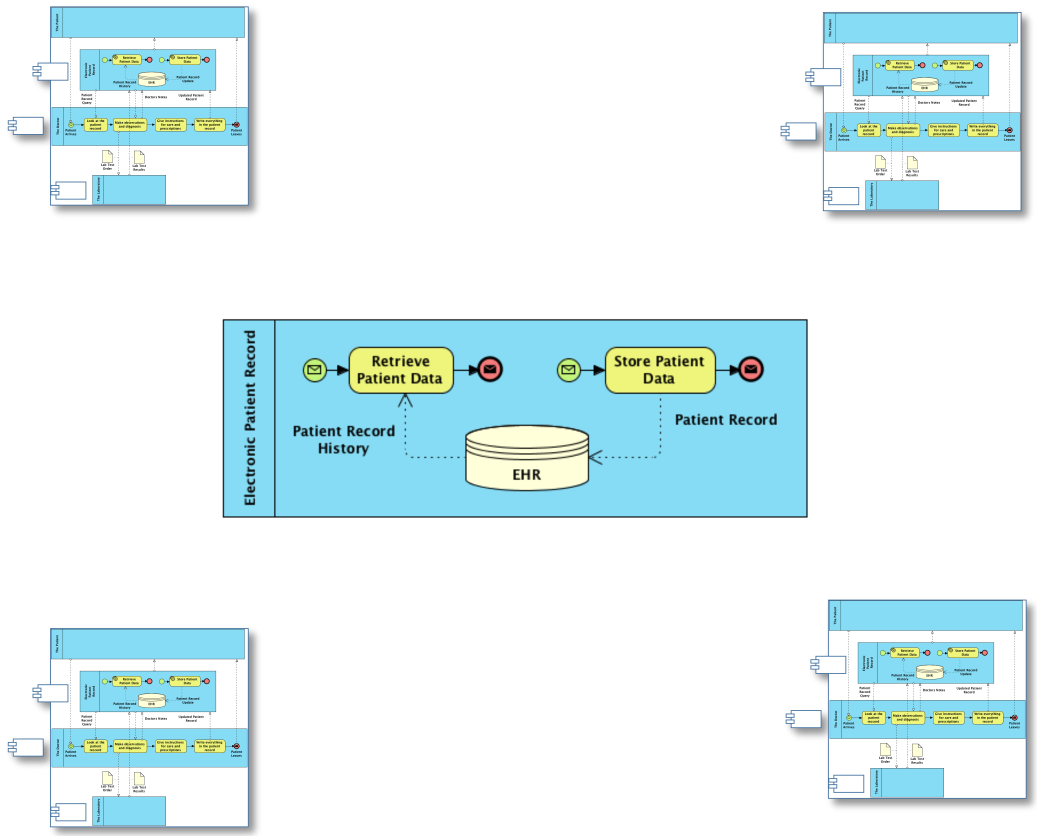
A standard for clinical documents

- HL7 CDA (Clinical Document Architecture) is a RIM based standard for exchange of clinical records
- EHR service can have the CDA documents as the payload in the messages



Extending the local EHR into regional EHR

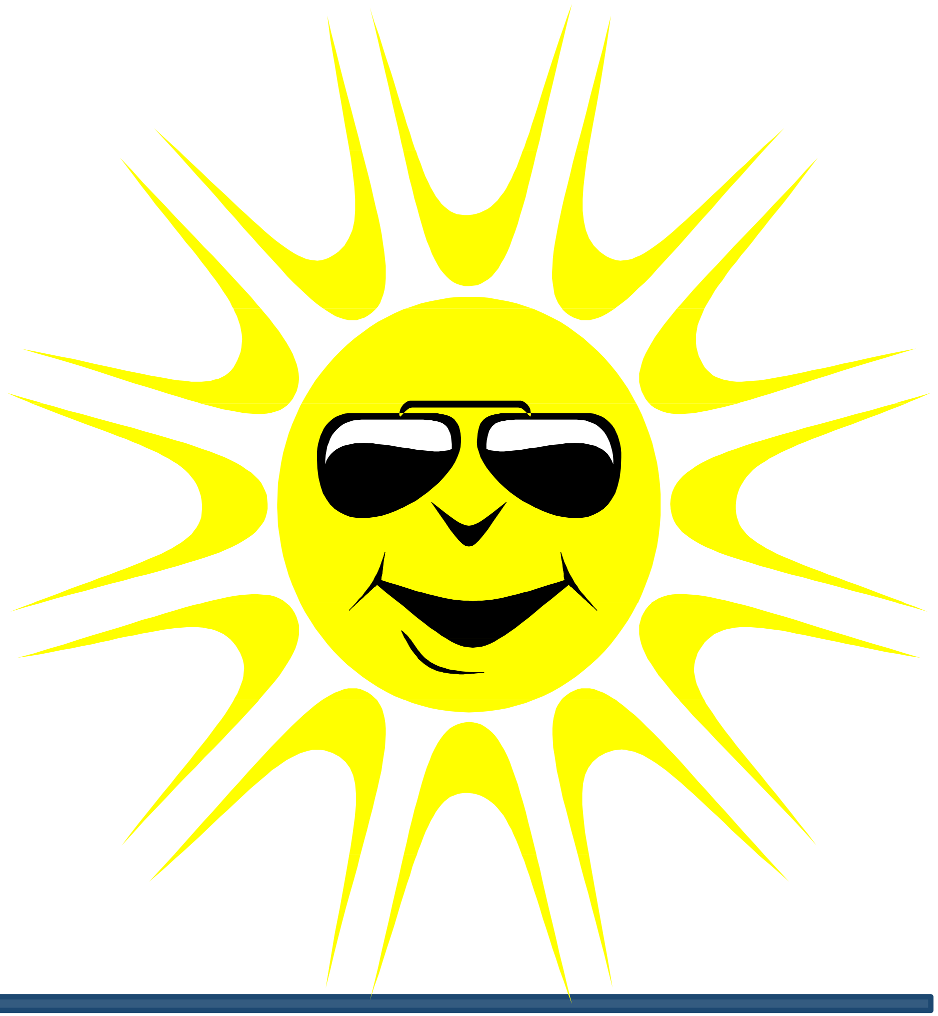
A Regional EHR service could help in data exchange between hospitals



Summary: How BPMN helps in improving the processes

- Modeling the As-Is business processes
- Identifying areas of improvement
- Discovering reusable business services
- Modeling the To-Be business processes
- Discovering web services
- Helping in implementation of the web services

Questions?



Contact
timo.itala(at)tkk.fi